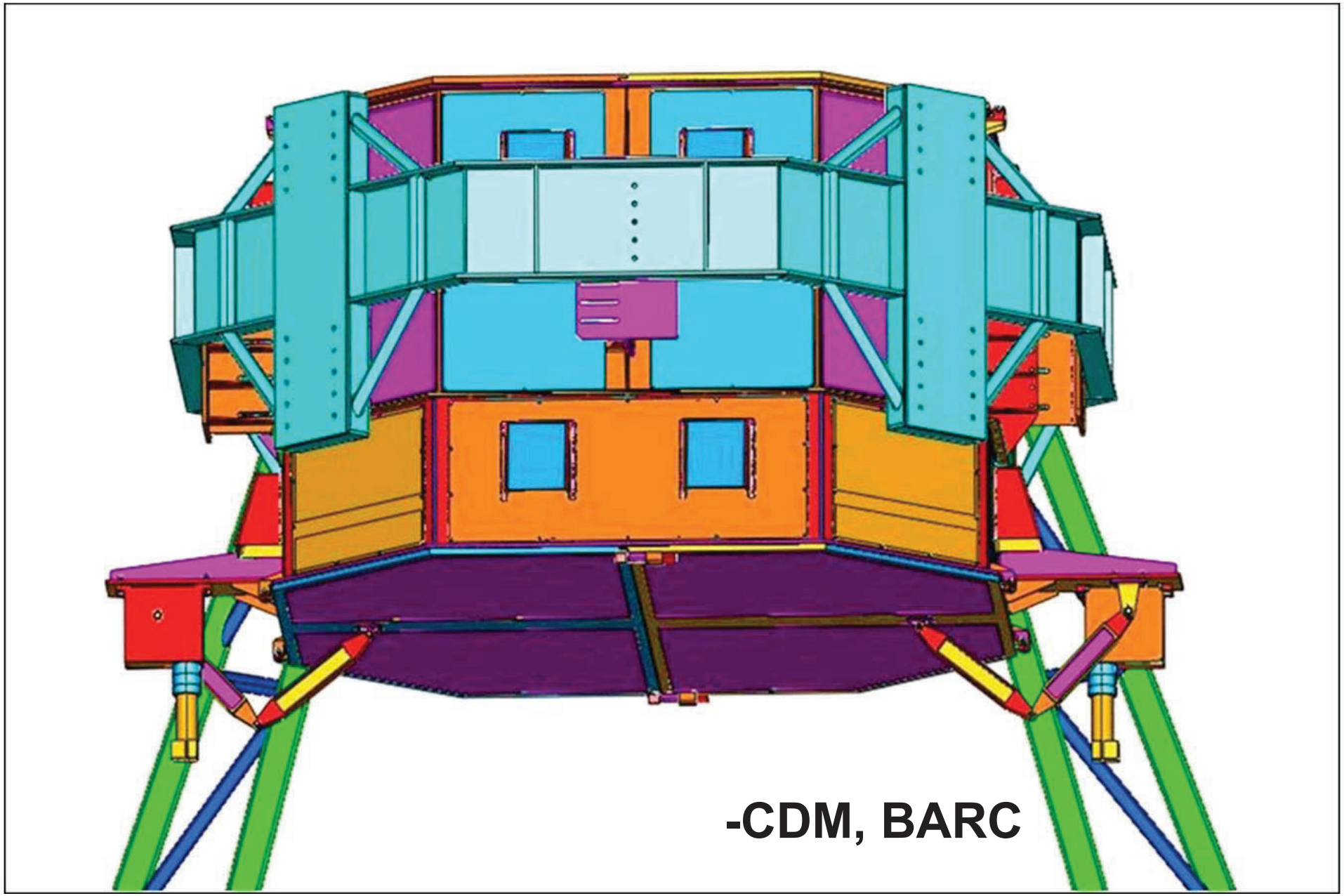
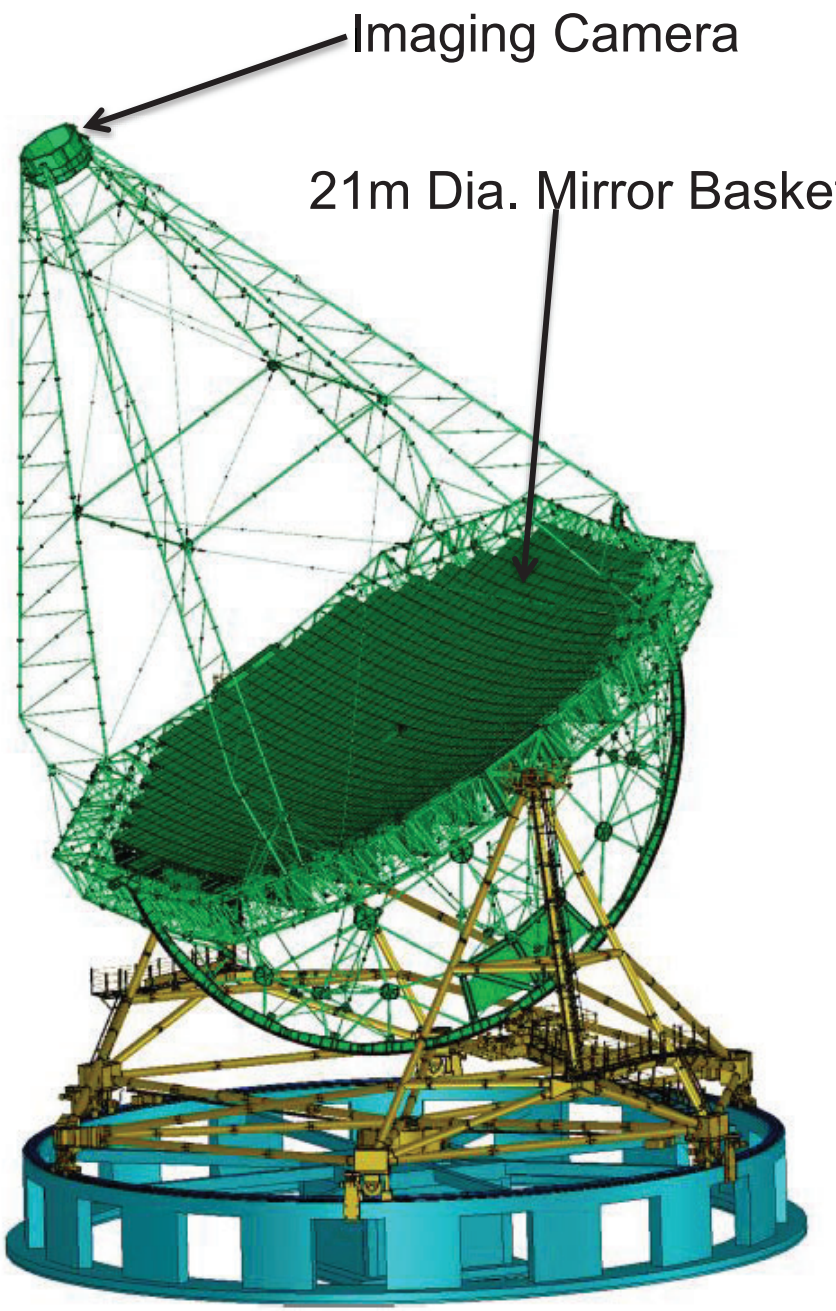
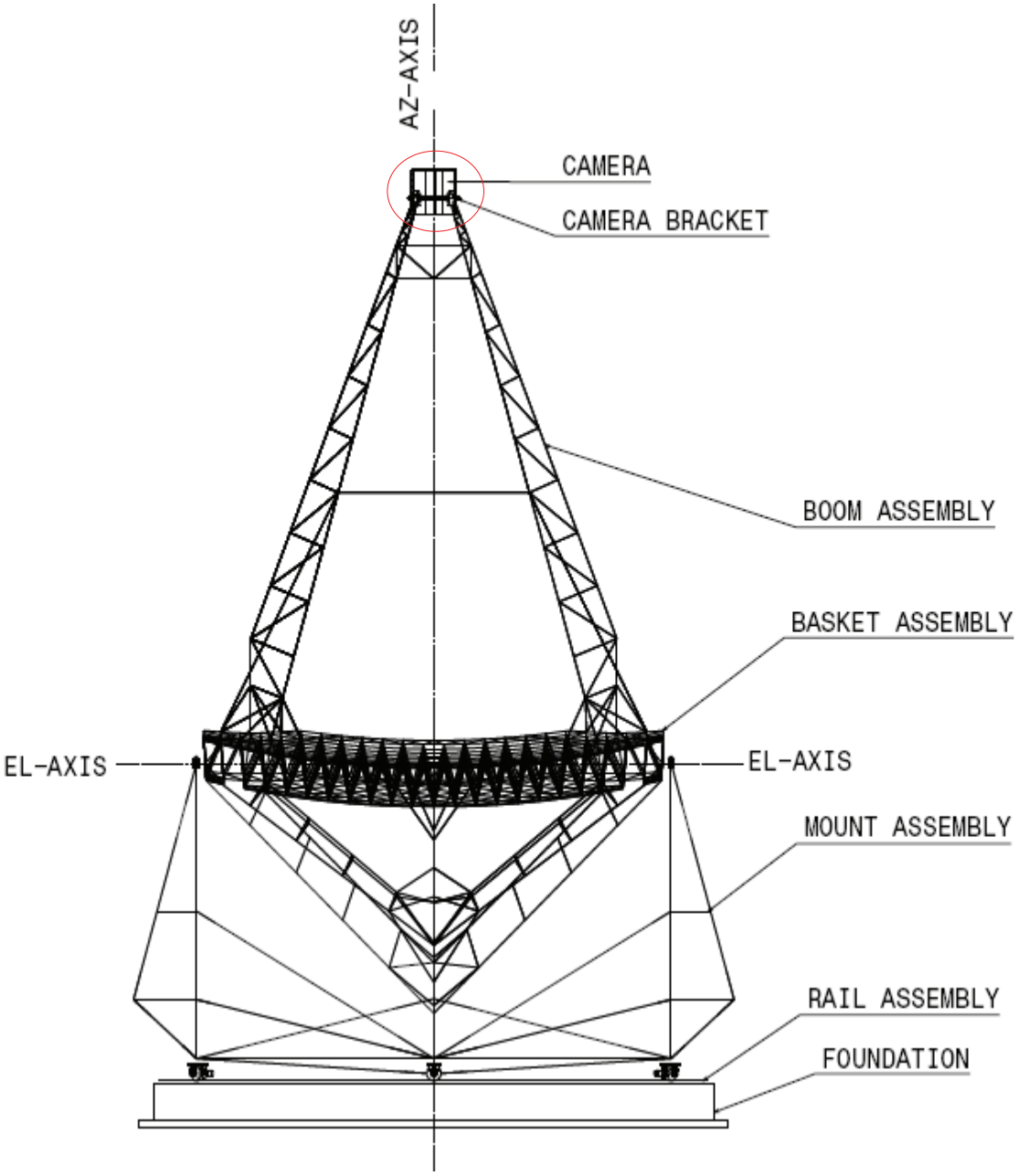
Development of Camera Housing for MACE

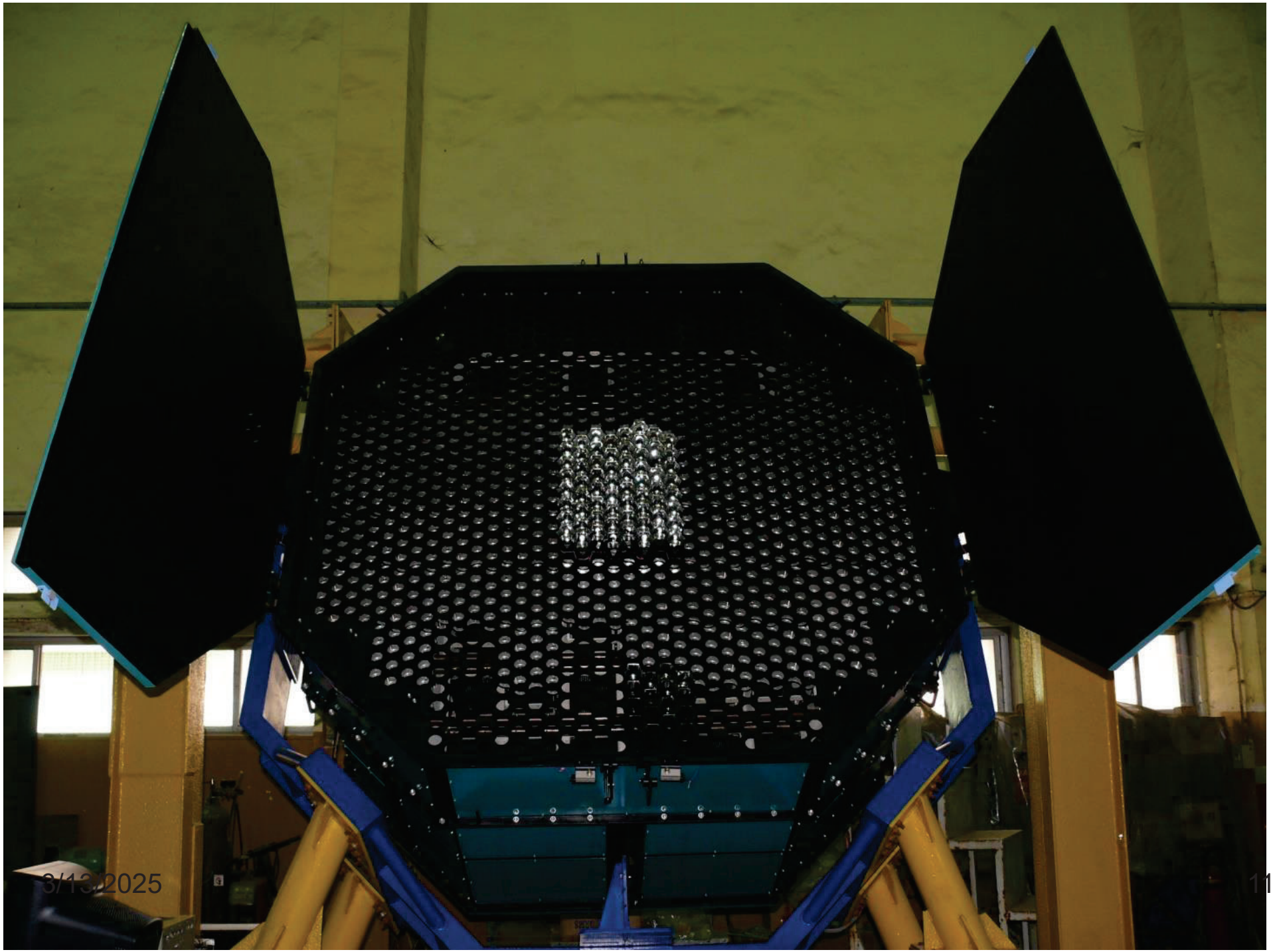




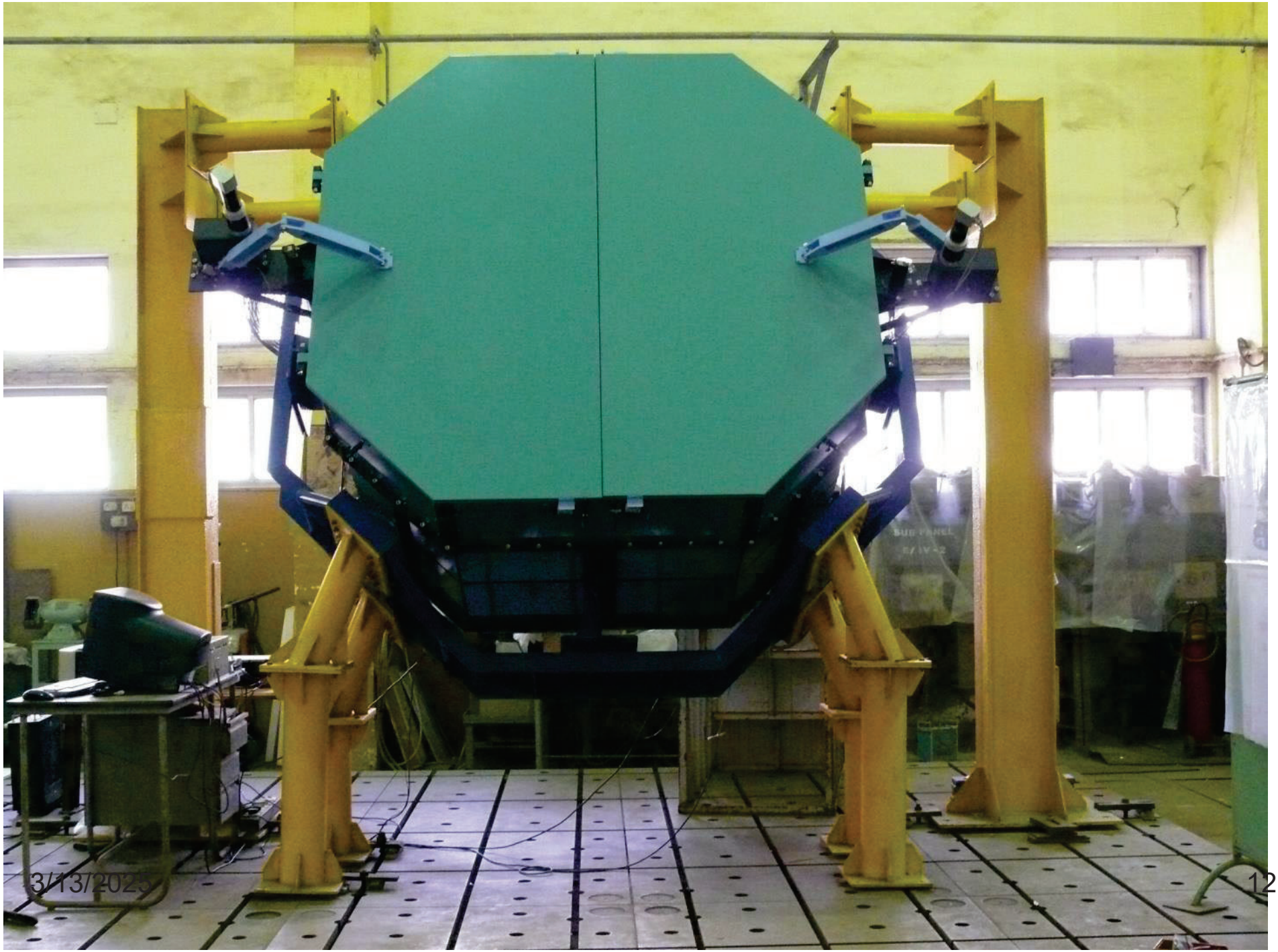


MACE Telescope

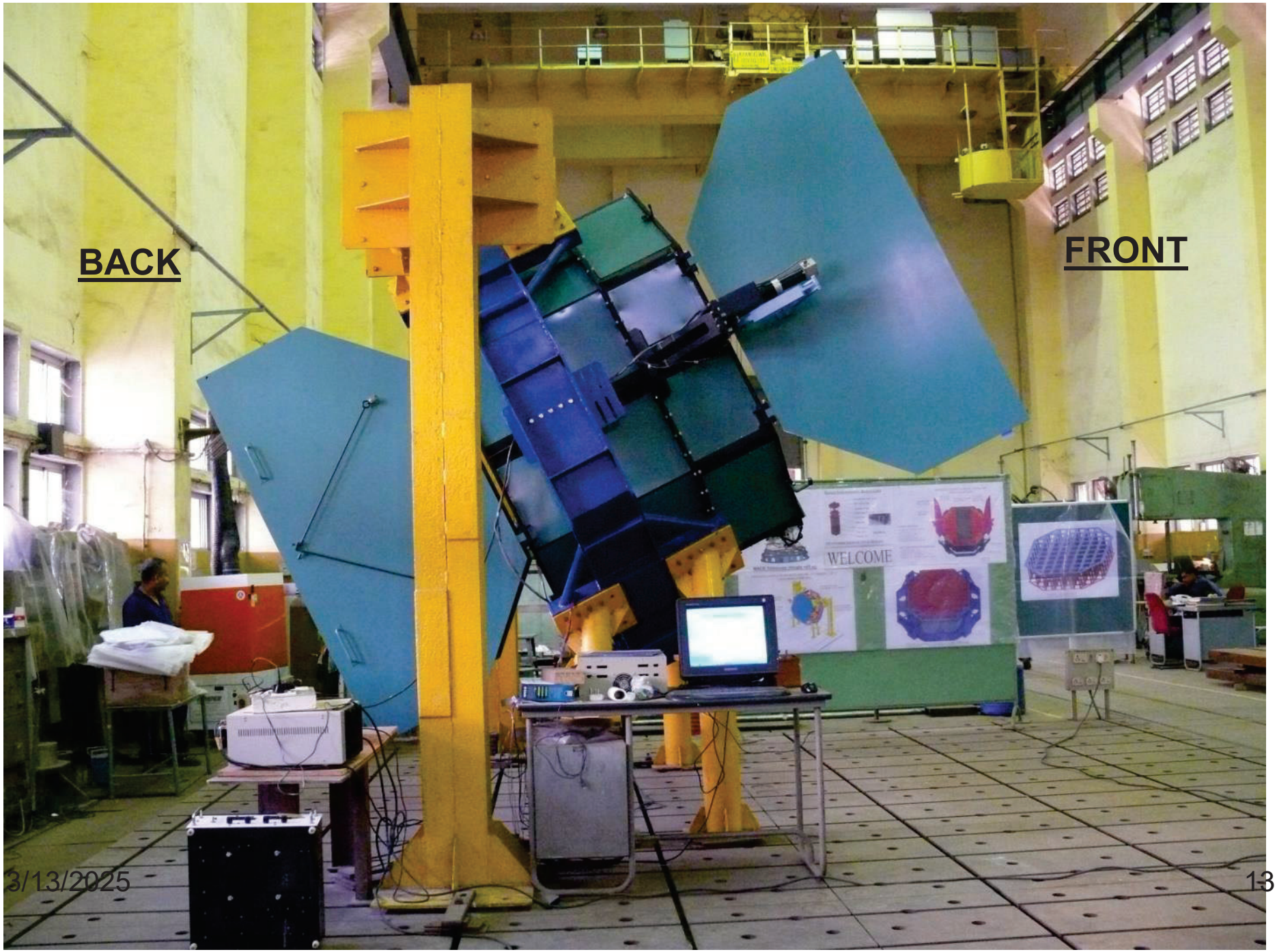
MACE Camera (front shutters open)



MACE Camera (front shutters closed)



MACE Camera (side view, shutters open)



1. INTRODUCTION

The Stereo MACE system consists of two Gamma ray telescopes, located a short distance apart, observing the same celestial target. The description in this chapter is pertaining to a single telescope. Thus, for two telescopes the quantities mentioned shall be double.

1.1. Description of System

Mace telescope has a camera mounted on the boom bracket at focal point of mirror surface which is around 25m as shown in Fig.1. The camera consists of Camera Integrated Modules (CIMs), which function as detectors for telescope, mounted in Camera Housing. This Camera Housing is further mounted on boom bracket, thereby it integrates with MACE telescope. The CIMs are arranged in 10x11 matrix on Camera Housing along with signal processing and data acquisition electronics, wiring and other axillary & support systems required for the operation and maintenance of the Camera. GA drawing of tentative design is attached as Annexure-1 for reference.

The Camera Housing shall be weather proof, light proof housing that is made of light materials yet stiff enough to safely house the system. The weight of the camera significantly impacts the load on the Mace structure. Hence, the camera weight must be minimised significantly to reduce the overall structural weight of the telescope. For this reason, Ti Grade-2 (UNS No.: R50400) is used as structural material and Al 6061-T6 is used to house electronics in CIM, port closure plates and other non-load bearing element in this referral design.

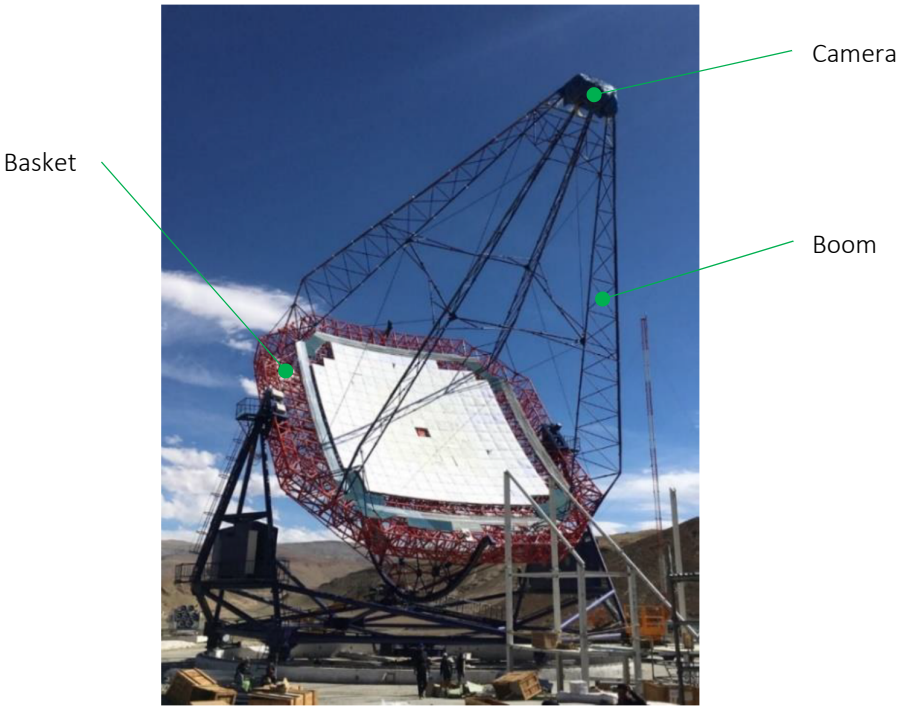


Figure 1. Snapshot of MACE Telescope at Hanle, Leh

Camera Housing has Front shutter and Back Door. During normal parking condition, both doors shall be in closed condition. For data collection during observations, the front shutter has to be remained in open condition for mirror reflected light to be detected by CIMs. Opening of front shutter has to be automated and operable in any orientation of telescope. For maintenance/inspection, the back doors shall be manually opened to get access to back side electronics. Ports (made of Al) are provided in shell for mounting electronics, other axillary & support systems required for the operation

and maintenance of the system and for cable routing etc. Heaters are provided in CPC shell to maintain optimum working temperature for camera electronics.

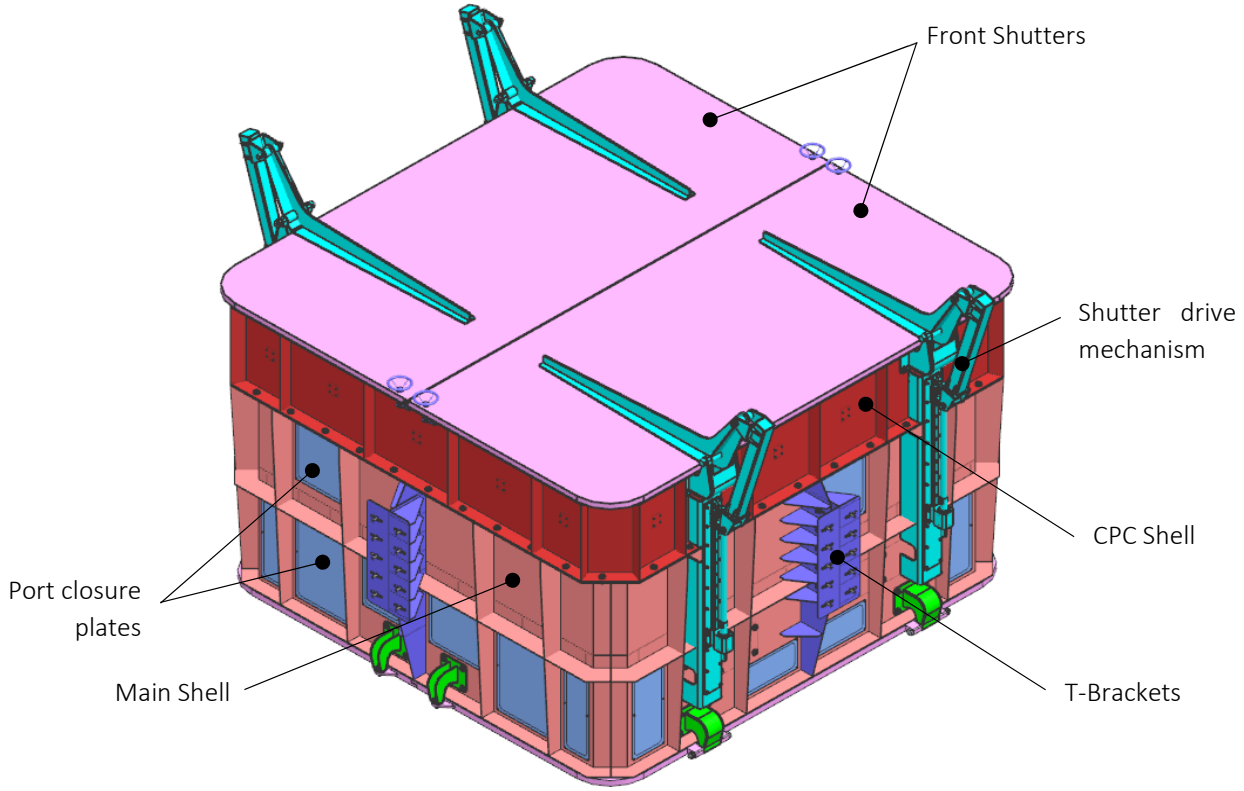


Figure 2. Solid Model of Stereo MACE Camera Assembly

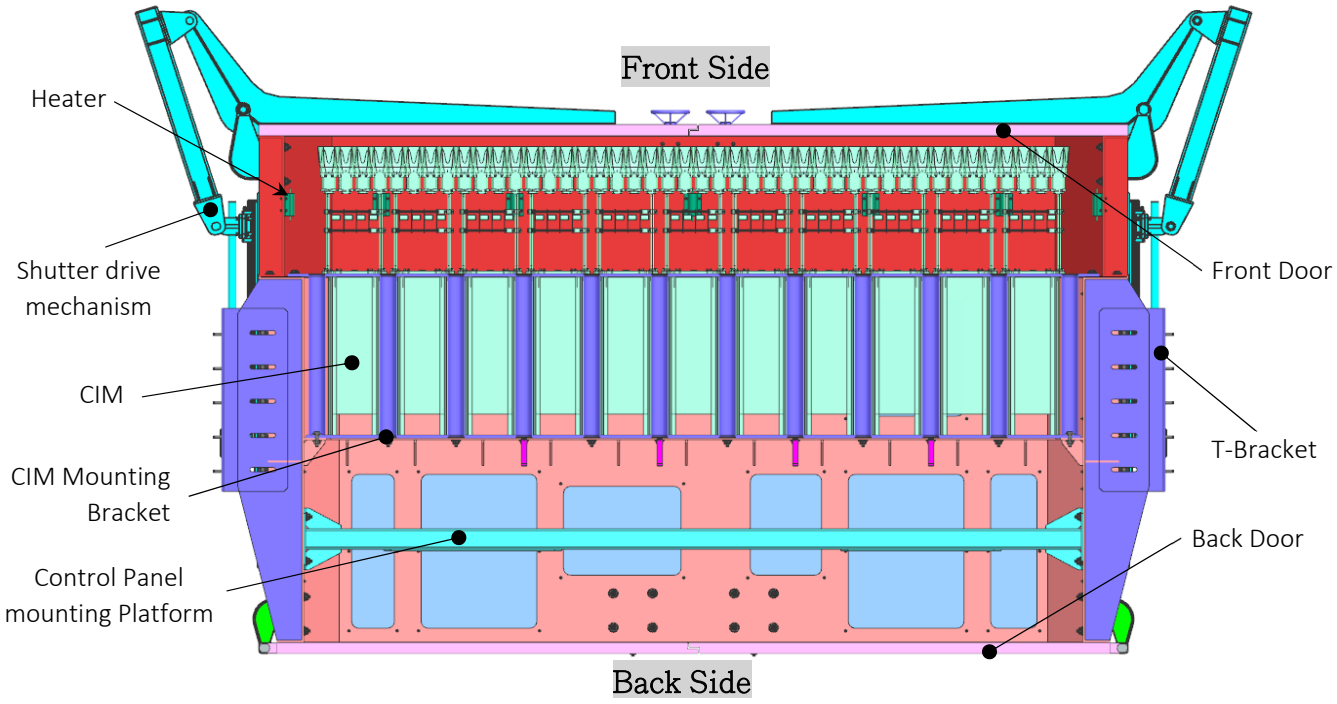


Figure 3. Front Sectional View of Stereo MACE Camera